Deep Learning for Image Denoising

From Denoising Autoencoders to Residual U-Net on CIFAR-10

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1 Introduction

Image denoising consists in reconstructing a clean image x from a corrupted observation $\tilde{x}$. It is a fundamental inverse problem in computer vision and signal processing.

In recent years, deep learning methods have significantly improved denoising performance compared to classical convolution-based filters. This project investigates convolutional neural networks for denoising natural images from the CIFAR-10 dataset.

We started from the principle of Denoising Autoencoders (Vincent et al., 2008) and progressively improved the architecture toward a residual U-Net with perceptual loss.

The objectives of this project are:

- Implement and evaluate deep denoising models with 2 types of noise corruption (Gaussian and speckle).
- Compare deep learning models against classical convolutional filters.

2 Related Work

2.1 Denoising Autoencoders

Denoising Autoencoders were introduced by Vincent et al. (2008) as a method for learning robust representations by reconstructing clean inputs from corrupted versions.

Given a corruption process:

$$\tilde{x} \sim q_D(\tilde{x} \mid x),$$

the model is trained to minimize:

$$\mathcal{L}(\theta) = \mathbb{E}_{x,\tilde{x}} \| f_\theta(\tilde{x}) - x \|_2^2.$$

2.2 Convolutional Architectures for Denoising

Fully connected autoencoders do not scale well to images. Convolutional networks exploit spatial locality and weight sharing, leading to improved performance.

U-Net architectures further improve reconstruction quality through multi-scale representations and skip connections.

2.3 Perceptual Loss

Pixel-wise MSE often leads to overly smooth reconstructions. Perceptual losses use pretrained networks (e.g., VGG16) to measure feature-level similarity.

3 Problem Formulation

We consider supervised image denoising:

$$\hat{x} = f_\theta(\tilde{x})$$

Two noise models are considered:

3.1 Gaussian Noise

$$\tilde{x} = x + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, \sigma^2)$$

3.2 Speckle Noise

$$\tilde{x} = x + x \odot \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, \sigma^2)$$

Images are normalized in $[0, 1]$.

4 Dataset

We use CIFAR-10, which consists of:

- 50,000 training images and 10,000 test images
- Resolution: 32×32 and 3 RGB channels

Noise is applied on-the-fly during training. For evaluation, noise generation is deterministic to ensure reproducibility.

5 Model Architecture : Residual U-Net

The final architecture is a lightweight 3-level U-Net.

Key features:

- Multi-scale encoder-decoder structure
- Skip connections
- Group Normalization
- Residual learning: the network predicts noise residual

Residual prediction:

$$\hat{x} = \text{clip}(\tilde{x} + r_\theta(\tilde{x}))$$

6 Training Procedure

6.1 Loss Function

The total loss combines:

- Pixel MSE
- Feature loss (VGG16 intermediate layer)
- Style loss (Gram matrix)

$$\mathcal{L} = \lambda_{mse}\mathcal{L}_{mse} + \lambda_{feat}\mathcal{L}_{feature} + \lambda_{style}\mathcal{L}_{style}$$

6.2 Optimization

- Optimizer: AdamW
- Learning rate: 3×10^{-3}
- Scheduler: Cosine Annealing
- Early stopping (patience = 10)
- Mixed precision training

7 Evaluation Metrics

7.1 Mean Squared Error (MSE)

$$MSE = \frac{1}{N}\sum_i (x_i - \hat{x}_i)^2$$

7.2 Peak Signal-to-Noise Ratio (PSNR)

$$PSNR = 10\log_{10}\left(\frac{1}{MSE}\right)$$

Higher PSNR indicates better reconstruction quality.

8 Results

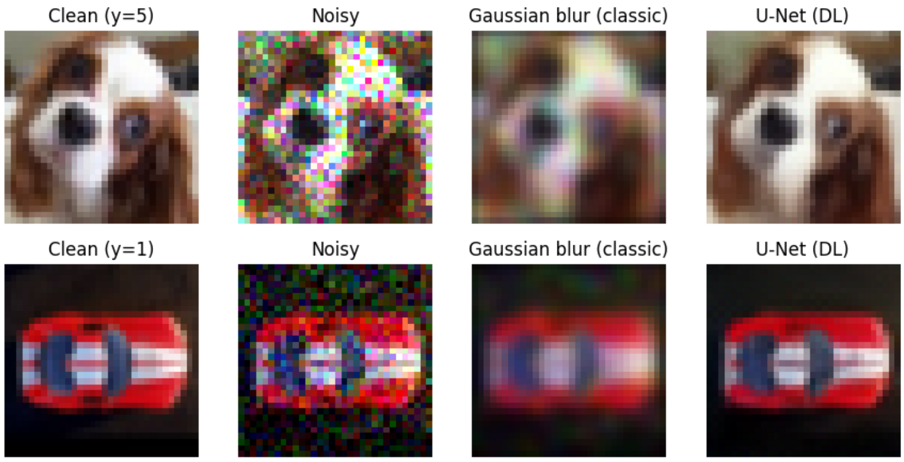
8.1 Quantitative Results

Noise Type	MSE	PSNR (dB)
Gaussian ($\sigma = 0.12$)	0.001899	27.22
Speckle ($\sigma = 0.35$)	0.002578	25.89

Table 1: Denoising performance for Gaussian and Speckle noise.

We can see that the model has a strong denoising effect, with an overall PSNR gain over identity of 9.58 dB. The Speckle noise remains more challenging than the Gaussian noise, but the model is able to retain strong performance even with the Speckle noise.

8.2 Comparison with Gaussian Blur



We can see that the model consistantly gives way better results than the gaussian filter, and overall really good qualitative results (some small details are lost, like the legs of the bird, but most of the colors and patterns are reconstructed without adding too much blur).

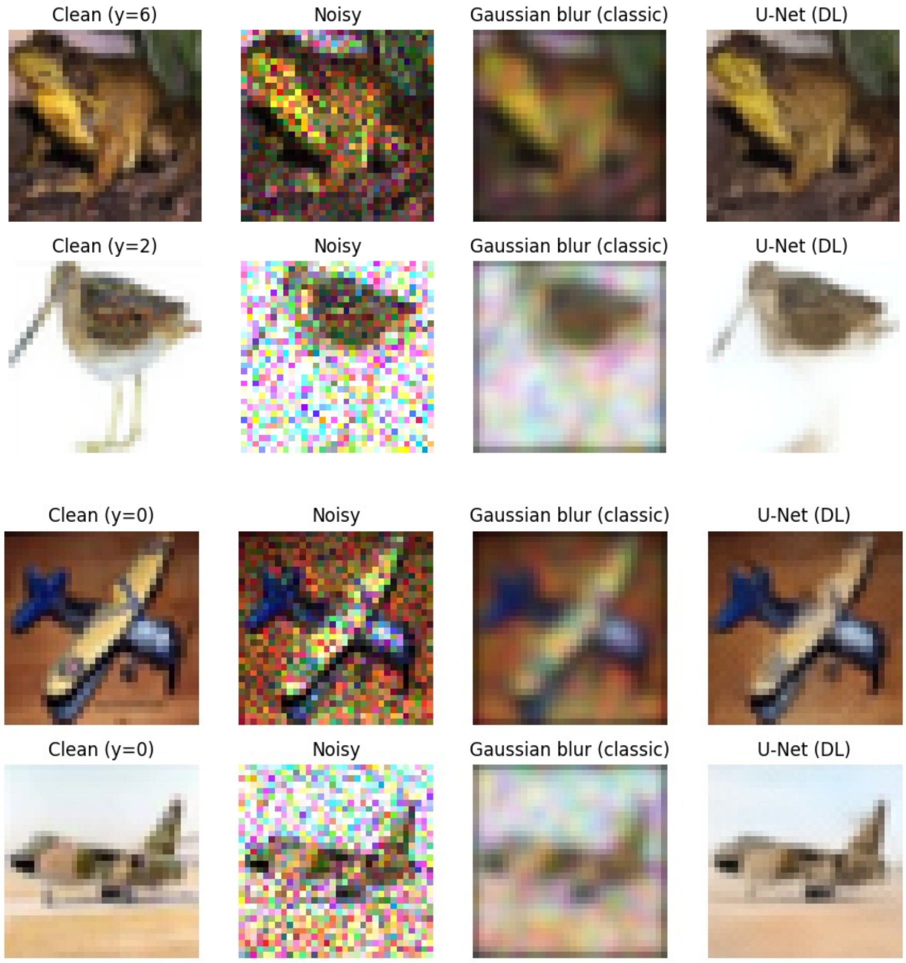


Figure 1: Examples of comparison between the original image, the image with noise, the image denoised with a classical gaussian filter, and the image denoised by my model

9 Conclusion

This project demonstrates that lightweight residual U-Nets with perceptual loss achieve strong denoising performance on CIFAR-10.

I wanted to explore more topics, but ran out of time / processing power :

- Larger images
- Analysis on the weights of each loss for the creterion
- Real-world noise modeling

References

- Vincent et al., Extracting and Composing Robust Features with Denoising Autoencoders, ICML 2008.
- Ronneberger et al., U-Net: Convolutional Networks for Biomedical Image Segmentation, 2015.
- Johnson et al., Perceptual Losses for Real-Time Style Transfer and Super-Resolution, 2016.